



02451 Introduction to Machine Learning

Week 10: K-means and hierarchical clustering

Georgios Arvanitidis

14 April 2026

DTU Compute, Technical University of Denmark

Today

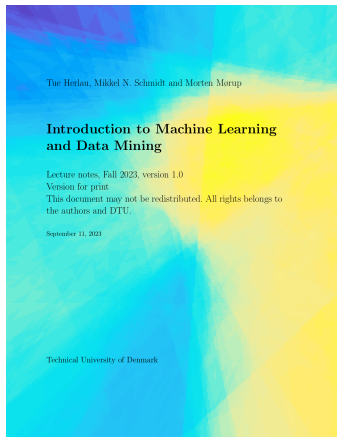
Feedback group of the day:

Olle Karl Oscar Åhrberg, Ayesha Aumran, Dragos Burlui, Patrick Byrne, Sara Cavar, Oliver Chaanhing, Mazarine Marie Chambon, Ho Lung Chu, Sarah Mikkelsen Dimovski, Mikkel Enevoldsen, Nehal Fernandes, Luis Sebastian Galán, Valdemar Galsgaard, Leah Gornstein, Gustav Rotne Hansen, Sebastian August Hjarndal, Naya Gutknecht-Stöhr Holst, Winston Harry Hueg, Harald Oiva Tapio Hvidlykke, Benjamin Iversen, Frederikke Louise Prytz Jensen, Asger Mechlenburg Johansen, Sherin Khalil, Jonas Rafn Kjær, Andreas Krarup, Anna Karolina Kulesza, Anastassia Levitseva, Lindsay Ma, Patrick Robert Anatol Machalet, Dlovan Mohamad, Frederik Nielsen, Sofie Marie Møller Nielsen, Niels Marcus Samuel Pilgaard, Benjamin Pedersen Rasmussen, Hans Frederik Sebber, Sebastian Theodor Sørensen, Lorenzo Stanghellini, Nitansh Thaker, Marios Topalis, Shikha Suketu Trivedi, Morten Schantz Vesth, Aleksandra Wozniak, Adal Tanac Yaman.

Reading/homework material:

Chapter 18

P18.1, P18.2, P18.3



Lecture Schedule

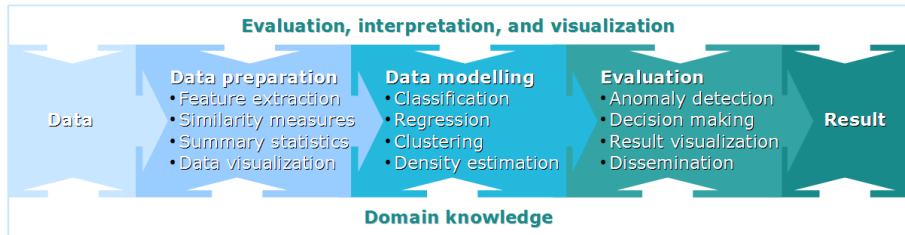
- 1 Introduction, data, and visualization
3 February: C1, C2, C7
- 2 Summary statistics, similarity, and nearest neighbors
10 February: C4, C12
- 3 Computational linear algebra and PCA
17 February: C3
- 4 Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5 Bayesian inference and linear models
3 March: C5, C6, C8
- 6 Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7 Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8 Artificial Neural Networks and Optimization
24 March: C15
- 9 Bias/Variance and ensemble methods
7 April: C14, 17
- 10 **K-means and hierarchical clustering**
14 April: C18 (Assignment 7-9 due on 16 April)
- 11 Mixture models and density estimation
21 April: C19, C20
- 12 Representation learning
28 April: Material provided separately
- 13 Recap and discussion of the exam
5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Learning Objectives



Learning Objectives

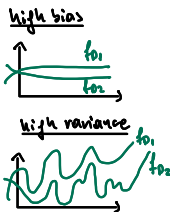
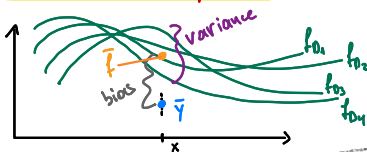
- Understand the principles behind K-means and hierarchical clustering
- Understand how different linkage functions affects clustering types
- Compare clustering solutions using Rand index, Jaccard and NMI
- Evaluate clustering quality using class labels

Practicalities and announcements

- Assignments 7-9 due to 16 April.
- Assignments 4-6 feedback underway.

Recap of Lecture 9

Bias-Variance decomposition



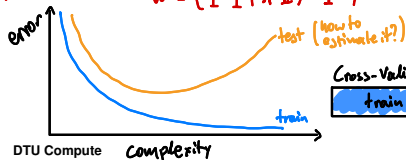
$$E_D[E^{f_{\text{est}}}] = E_x[\underbrace{\text{Var}_{y|x}[Y]}_{\text{variance}} + \underbrace{(\bar{y}(x) - f(x))^2}_{\text{bias}} + \underbrace{\text{Var}_D[f_0(x)]}_{\text{variance}}]$$

Suppose $f(x; w) = w^T \phi(x) = w_0 + w_1 \cdot \phi_1(x) + w_2 \cdot \phi_2(x) + \dots$

for example $\phi(x) = [1, x, x^2, x^3, x^4, \dots, x^k]$ for $k \leq N$

Regularized loss $L(w) = \frac{1}{N} \sum_{i=1}^N (y_i - f(x_i; w))^2 + \lambda \cdot \|\bar{w}\|_2^2$

Optimal solution $w^* = (\Phi^T \Phi + \lambda \cdot \bar{I})^{-1} \cdot \Phi^T \cdot y$ [$w_1, \dots, w_k$]



Ensemble Methods

- We consider a weak classifier (logistic regression) and we want to make it powerful
- "Cradle" k datasets by subsampling (with replacement)
- Bagging:
 - prob to sample a point the same for all always
 - predict with majority voting.

training data D

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

D_1

1	3	3	7	1
---	---	---	---	---

 $\dots$ D_k

2	5	1	8	4
---	---	---	---	---

- Boosting:
 - prob to sample a point changes over the training
 - predict with weighted majority voting.

D_1

5	8	1	9	7	4	5	2
---	---	---	---	---	---	---	---

$\downarrow M_1$

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

 Increase the probability to sample the ones predicted wrong.

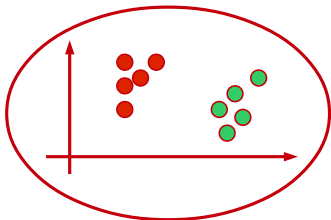
D_2

8	4	1	3	4	1	6	7
---	---	---	---	---	---	---	---

Remarks

- We can use 2-level CV (5-outer, 10-inner) and use the selected 5 models in an ensemble.
- We can use strong models in an ensemble. This is already a set of powerful models. In bagging, this can reduce the variance that comes from the complexity of the strong model.

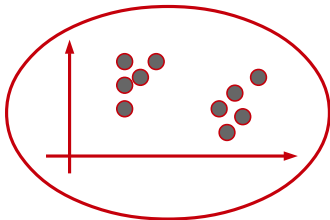
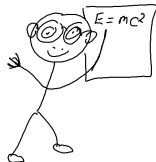
Supervised and Unsupervised learning



Supervised Learning

Input data $\mathbf{x}_n$ and output y_n

(Classification and Regression)



Unsupervised Learning

Input data $\mathbf{x}_n$ alone

(Exploratory analysis)



Imagine you observe the world for the first time!

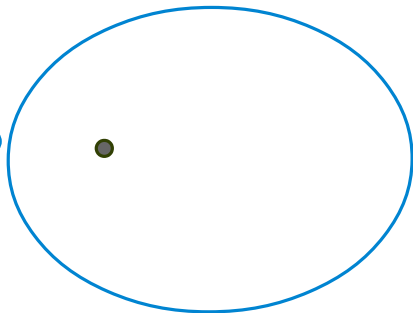


<http://www.clipartlord.com/category/baby-clip-art/>

Imagine you observe the world for the first time!



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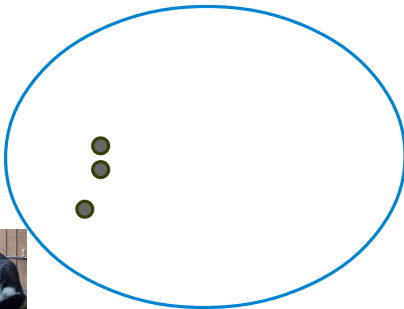
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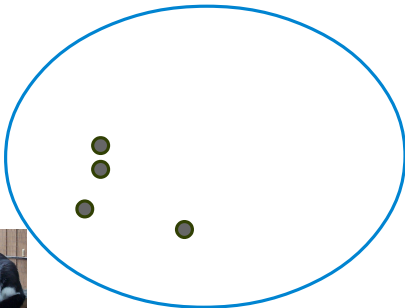
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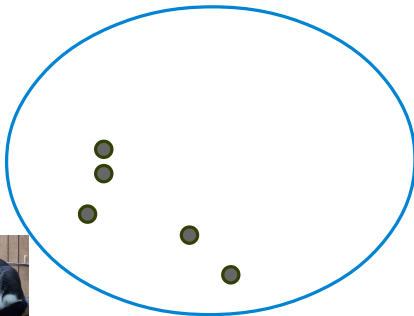
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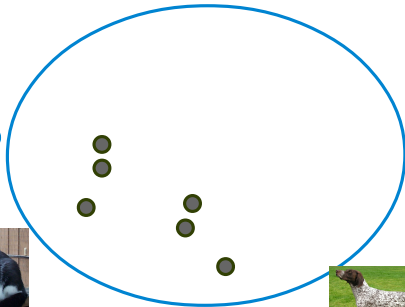
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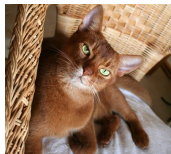
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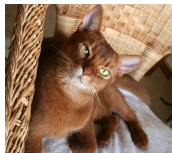
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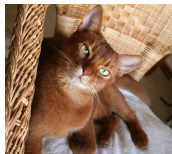
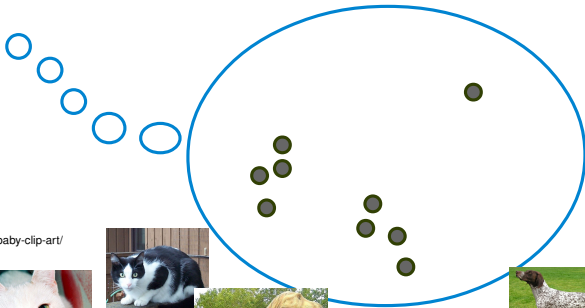
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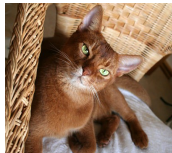
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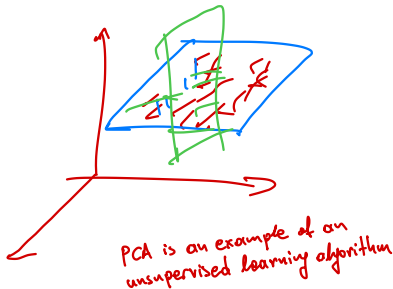


We humans are skilled at dividing objects into groups (clustering), but how do we make computers do the same?

Source (animal images): commons.wikimedia.org

Unsupervised learning

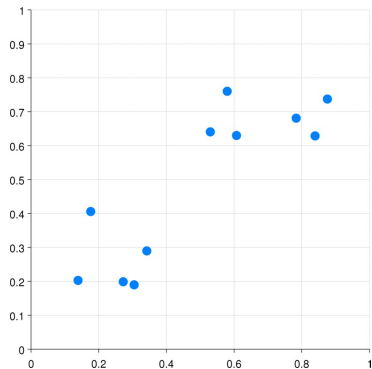
- Supervised learning
 - Use the data to learn a mapping from input to output that aligns with the purpose/task
- Unsupervised learning
 - No output variables available
 - Sometimes called exploratory analysis
 - What to learn from the data?
 - Structure
 - Regularities
 - Hidden information
 - Etc.



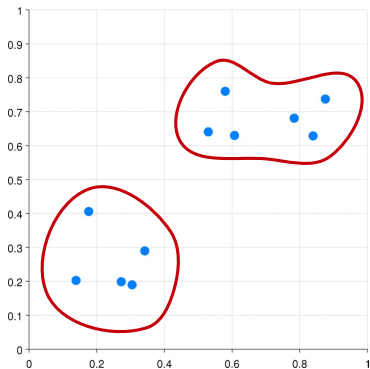
Clustering

- Divide data into groups (subsets/clusters) that are
 - **Meaningful:** Capture the natural structure of the data provided
 - **Useful:** Depends on purpose/task
- Observations in the same cluster are **similar in some sense**
- Unsupervised classification

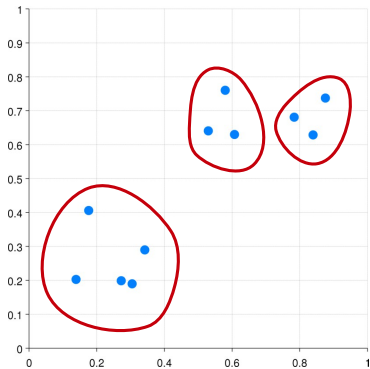
Clustering



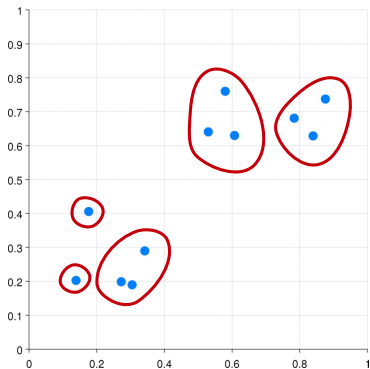
Clustering



Clustering

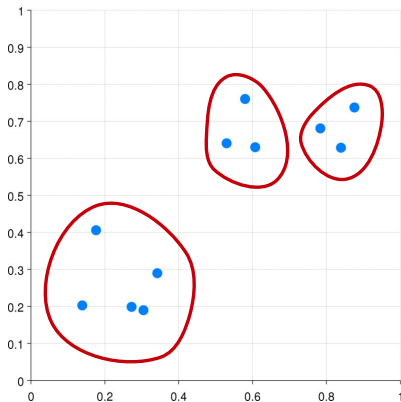


Clustering

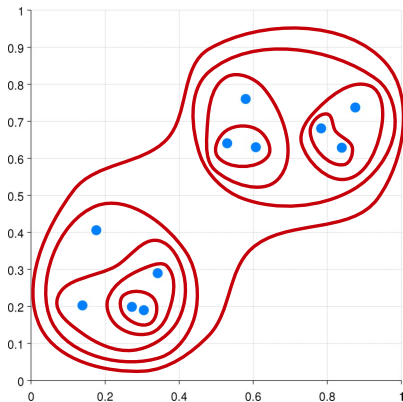


Partitional / hierarchical clustering

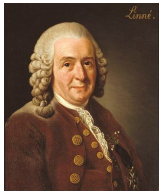
Partitional



Hierarchical



Phylogenetic trees



Carl Linnaeus
(1707 – 1778)

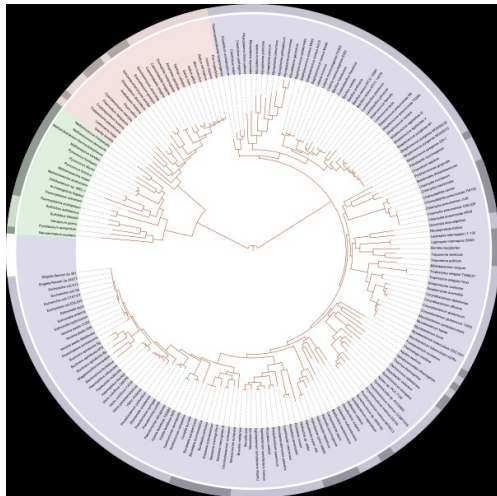


Image source:

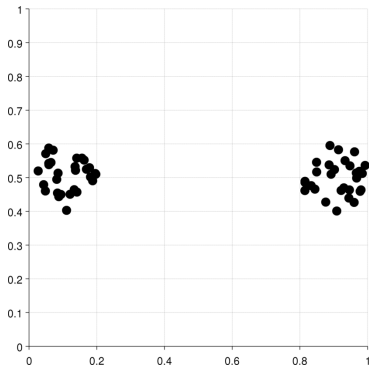
http://en.wikipedia.org/wiki/Carl_Linnaeus

http://en.wikipedia.org/wiki/File:Tree_of_life_SVG.svg

Types of clustering

Well-separated

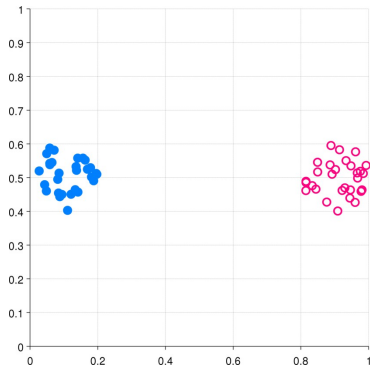
- Each point is closer to all points in its cluster than any point in another cluster.



Types of clustering

Well-separated

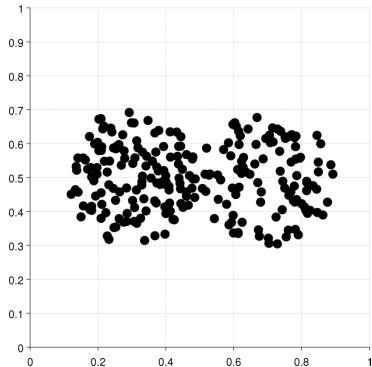
- Each point is closer to all points in its cluster than any point in another cluster.



Types of clustering

Center-based

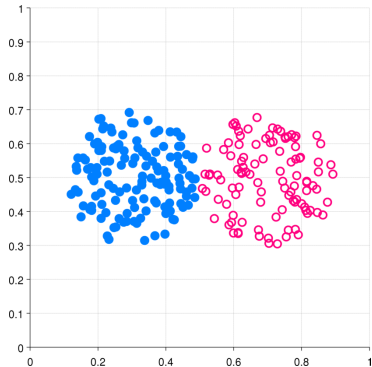
- Each point is closer to the center of its cluster than to the center of any other cluster



Types of clustering

Well-separated

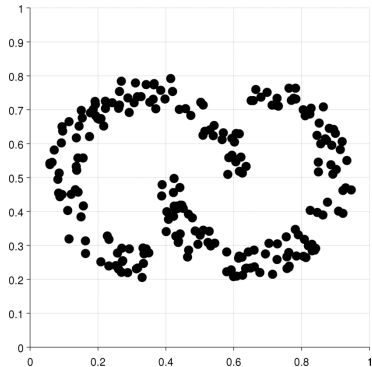
- Each point is closer to the center of its cluster than to the center of any other cluster



Types of clustering

Contiguity-based

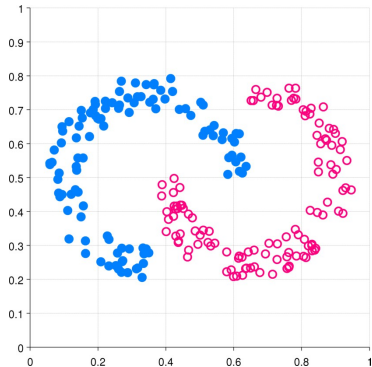
- Each point is closer to at least one point in its cluster than to any point in another cluster



Types of clustering

Contiguity-based

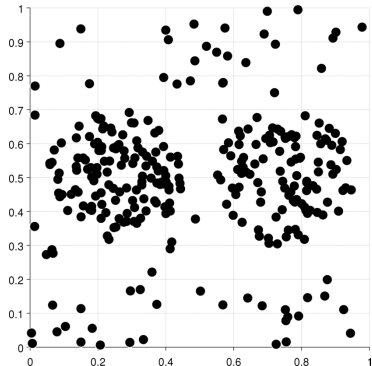
- Each point is closer to at least one point in its cluster than to any point in another cluster



Types of clustering

Density-based

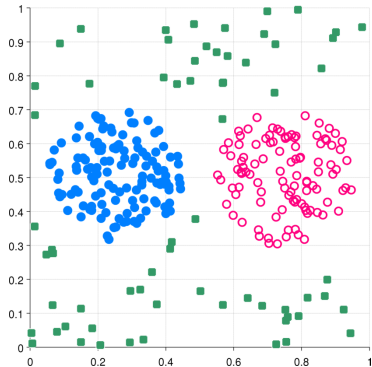
- Clusters are regions of high density separated by regions of low density



Types of clustering

Density-based

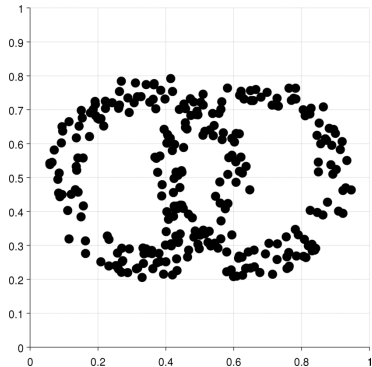
- Clusters are regions of high density separated by regions of low density



Types of clustering

Conceptual clusters

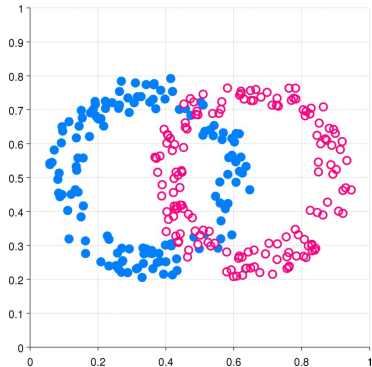
- Points in a cluster share some general property that derives from the entire set of points



Types of clustering

Conceptual clusters

- Points in a cluster share some general property that derives from the entire set of points



Quiz 01: Clustering types

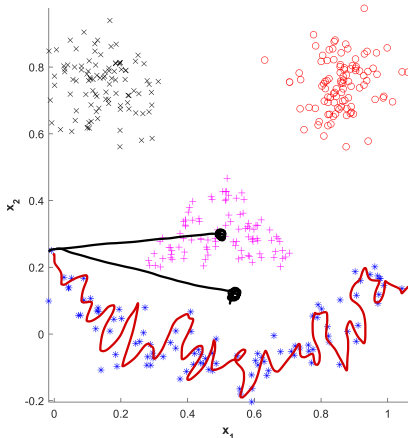


Figure 1: A clustering problem containing four clusters indicated by black crosses, red circles, magenta pluses and blue stars.

Consider the clustering problem given in Figure 1. Which clustering approach is *most* suited for correctly separating the data into the four groups indicated by black crosses, red circles, magenta pluses, and blue asterics?

- A. A well-separated clustering approach.
- ☒ B. A contiguity-based clustering approach.
- C. A center-based clustering approach.
- D. A conceptual clustering approach.
- E. Don't know.

Solution:

As the observation in each cluster is at least closest to one other observation in its cluster than to an

observation in another cluster a contiguity based approach is most suited.

K-means clustering

Select K points as initial centroids

Repeat

- Form K clusters by assigning each point to its closest centroid
- Recompute the centroids of each cluster

Until centroids do not change

Default distance function: The basic/classical K-means algorithm uses the squared Euclidean distance.

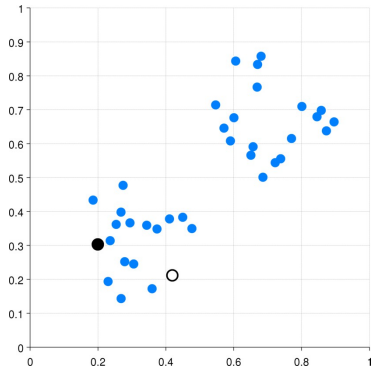
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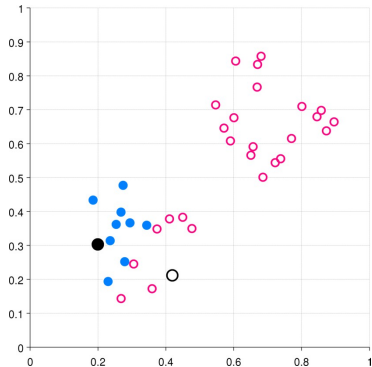
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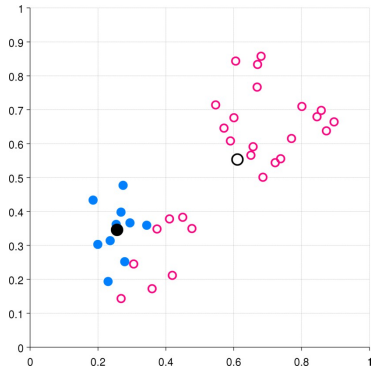
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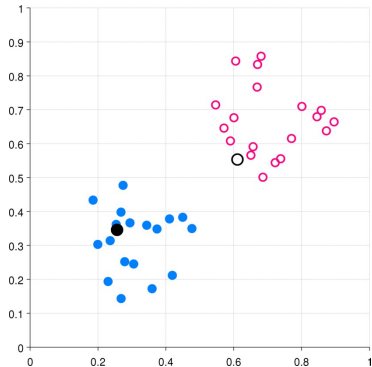
K-means clustering

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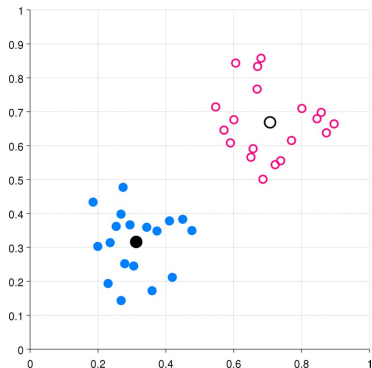
K-means clustering

Select K points as initial centroids

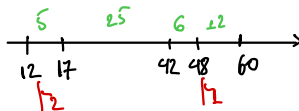
Repeat

- Form K clusters by assigning each point to its closest centroid
- Recompute the centroids of each cluster

Until centroids do not change



Quiz 02: K-means



Consider the following dataset

$$X = \{42, 60, 17, 48, 12\}$$

Select K points as initial centroids
Repeat

- Form K clusters by assigning each point to its closest centroid
- Recompute the centroids of each cluster

Until centroids do not change

We wish to apply the K -means algorithm with $K = 2$ clusters to this dataset and we initialize with cluster centroids at $\mu_1 = 17$ and $\mu_2 = 12$. Carefully, using pen and paper, go through each step of the K -means algorithm until it converge. What is the final clustering?

- A. $\{60, 48\}, \{12, 17, 42\}$
- B. $\{42, 60, 48, 17\}, \{12\}$
- C. $\{60\}, \{12, 17, 42, 48\}$
- ☒ D. $\{42, 60, 48\}, \{12, 17\}$
- E. Don't know.

Solution:

The correct answer is *D*. We will verify this by listing the intermediate steps of the *K*-means algorithm:

1. The initial clustering will be

$$\{42, 60, 48, 17\}, \quad \{12\}.$$

2. The new centroids will be

$$\mu_1 = \frac{42 + 60 + 48 + 17}{4} = 41.75, \mu_2 = \frac{12}{1} = 12.$$

3. The new clusters will then be

$$\{42, 60, 48\}, \quad \{12, 17\}.$$

4. The centroids then become:

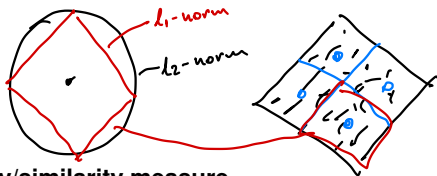
$$\mu_1 = \frac{42 + 60 + 48}{3} = 50, \mu_2 = \frac{12 + 17}{2} = 14.5.$$

It is easy to verify the cluster assignment/centroids will no longer be updated and the method therefore stops.

K-means clustering

How do I

- Find the closest centroid?
 - Use a suitable **dissimilarity/similarity measure**
- Compute the cluster centroids?
 - Depends on dissimilarity/similarity measure
 - For example, for squared Euclidean distance the mean is optimal
- Decide on K - the number of centroids?
- Initialize the K centroids?



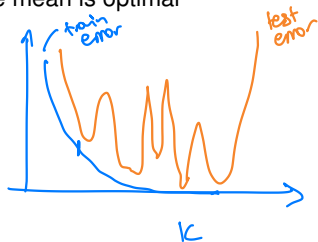
$$f_i^* = \arg \min_{f_i} \sum_{j \in G_i} d(f_i, x_j)$$

the points in the dataset for which f_i is the closest center

$d(f_i, x_j)$ the distance measure we selected

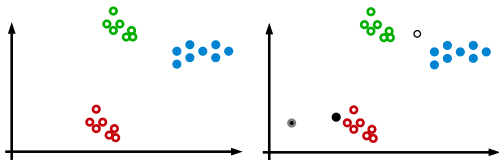
examples: $\|x - y\|_2$, $\|x - y\|_2^2$, $\|x - y\|_1$

each of them gives different f_i^* as optimal e.g. for $\|x - y\|_2^2$, the f_i^* is the mean of G_i

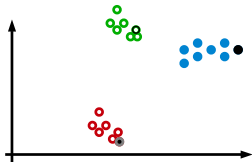


K-means clustering - initialization

- How will the data (top-left diagram) be clustered given the initialization of the three centroids shown at the right and at the bottom?



- What could we do if we have an empty cluster?
- What could be a good initialization procedure? (Farthest First)



The K-means solution depends on the initialization!

Agglomerative hierarchical clustering

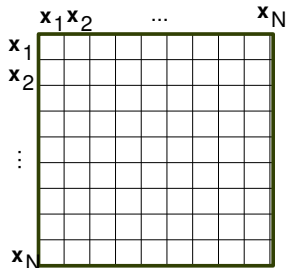
Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains

$$D_{ij} = \text{distance}(x_i, x_j)$$



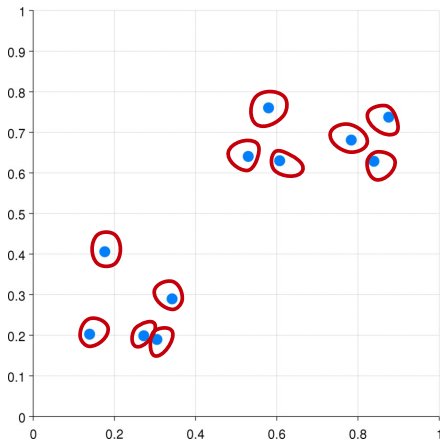
Agglomerative hierarchical clustering

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- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



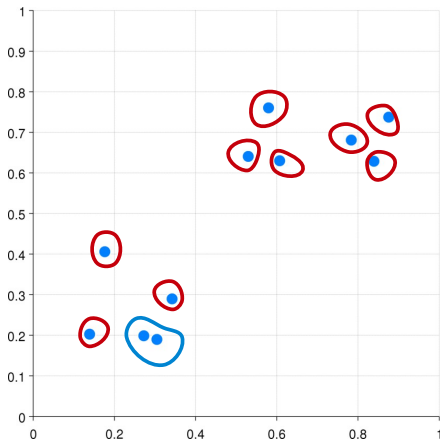
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



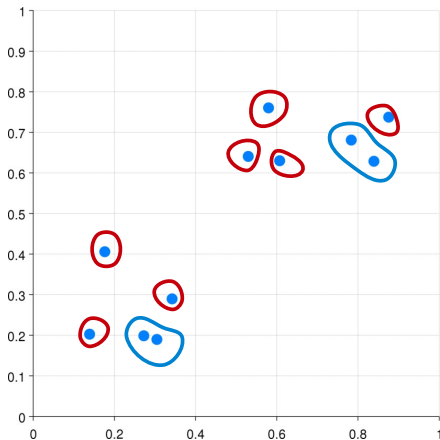
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



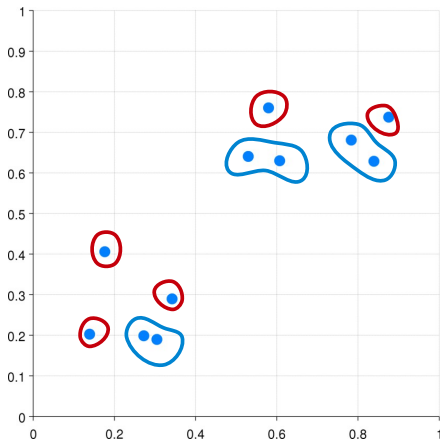
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



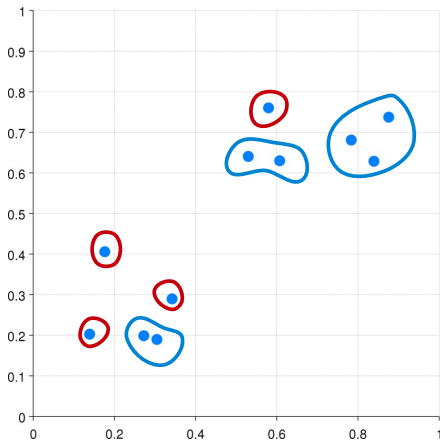
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



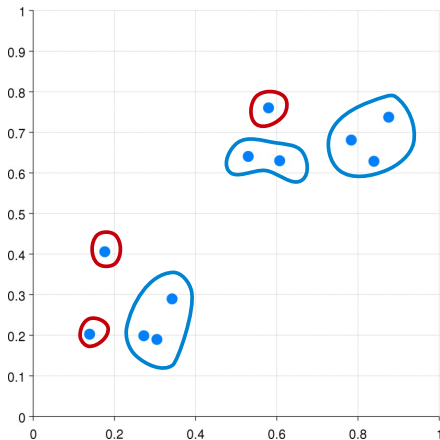
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



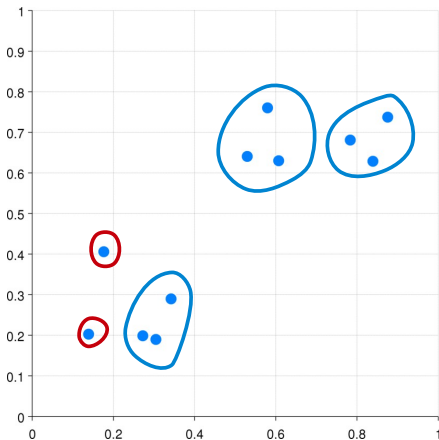
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



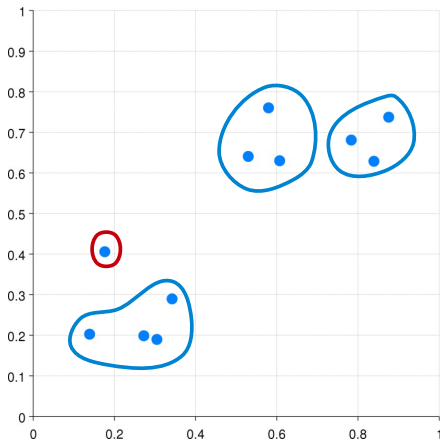
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



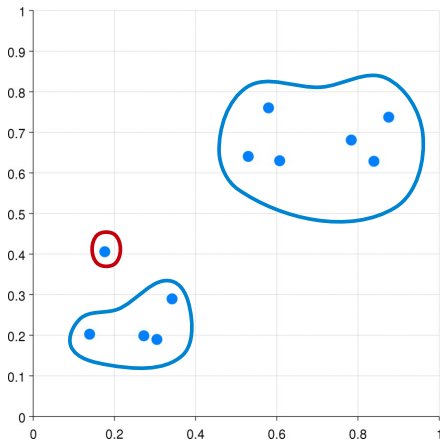
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



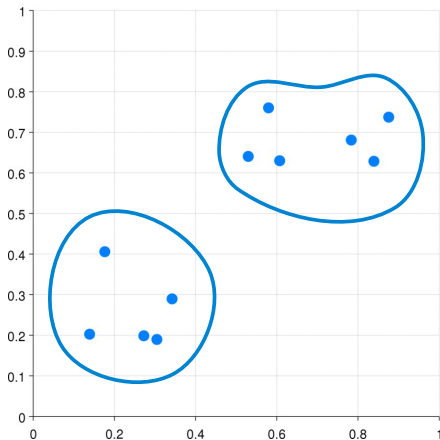
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



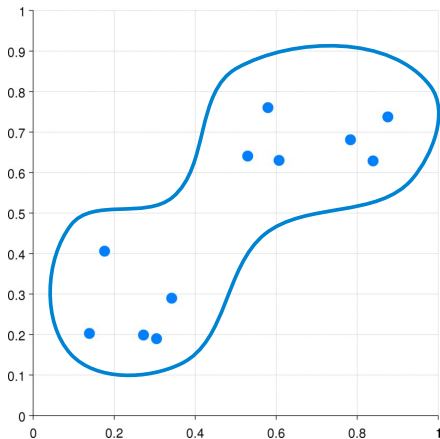
Agglomerative hierarchical clustering

Initialize the proximity matrix

Repeat

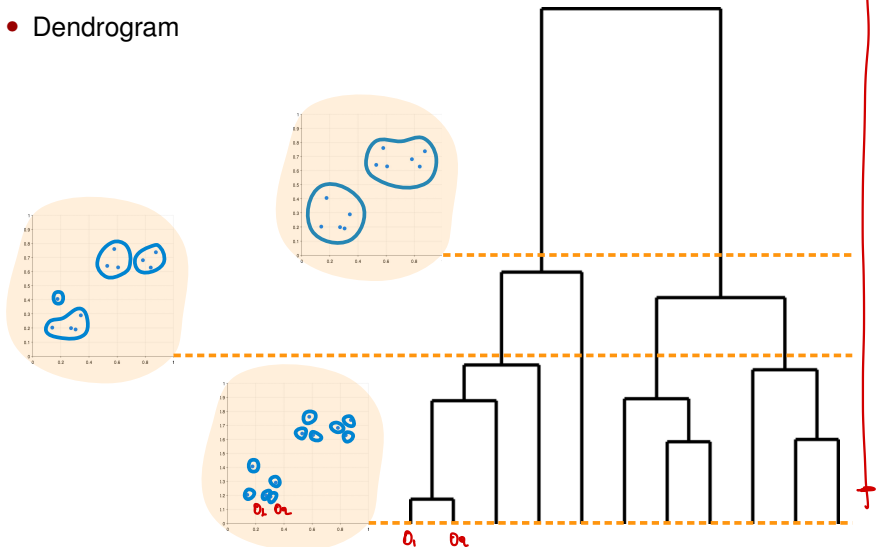
- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains



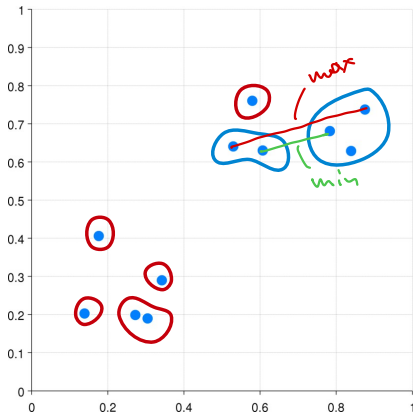
Agglomerative hierarchical clustering

- Dendrogram



Similarity between clusters

- The **key operation** in in agglomerative hierarchical clustering is measuring distance (dissimilarity) between clusters



Proximity between clusters

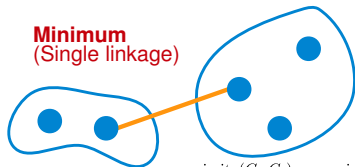
- Can be computed using **proximity between objects**
- In our example before we used Euclidian distance as proximity measure

C_i : Observations in cluster i

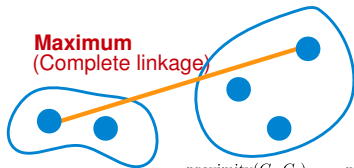
C_j : Observations in cluster j

m_i : Number of observations in cluster i

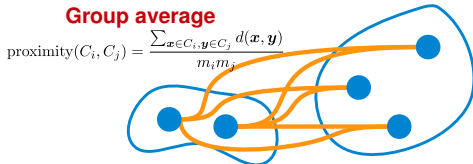
m_j : Number of observations in cluster j



$$\text{proximity}(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$



$$\text{proximity}(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$

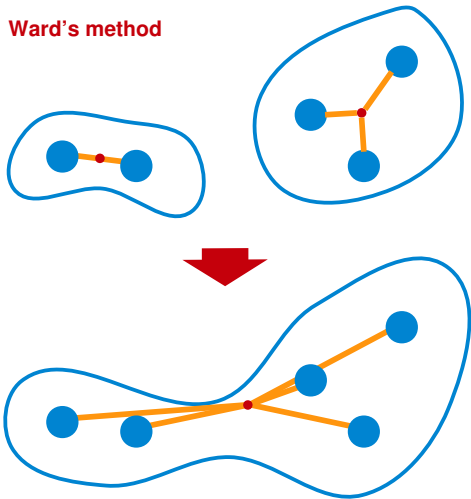


$$\text{proximity}(C_i, C_j) = \frac{\sum_{x \in C_i, y \in C_j} d(x, y)}{m_i m_j}$$

Proximity between clusters

- Increase in sum of squared error after merging the two clusters should be as small as possible

Ward's method



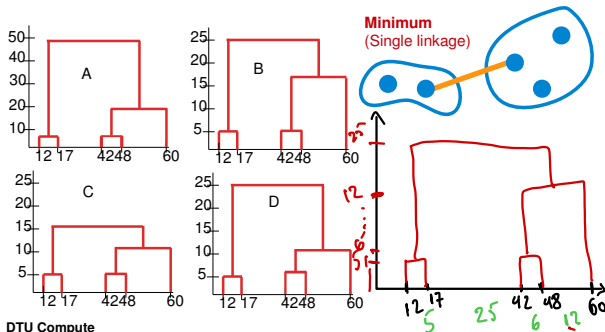
Quiz 03: Dendrograms

Consider once more the dataset:

$$X = \{42, 60, 17, 48, 12\}$$

Using pen-and-paper, carefully build a dendrogram from X one step at a time using Euclidean distance and *minimum* (single) linkage. What will the dendrogram look like?

- A. Dendrogram A
- B. Dendrogram B
- C. Dendrogram C
- D. Dendrogram D
- E. Don't know.



Initialize the proximity matrix
Repeat

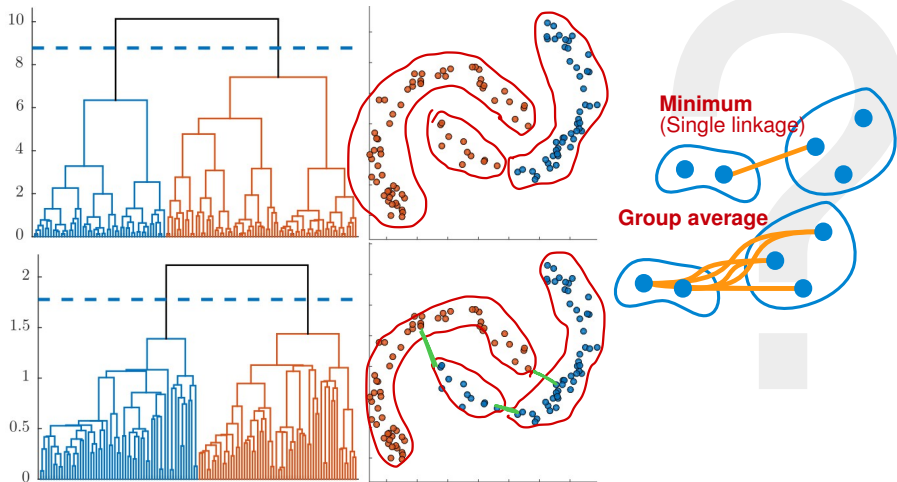
- Merge the two closest clusters
- Update the proximity matrix to reflect the proximity between the new cluster and the original clusters

Until only one cluster remains

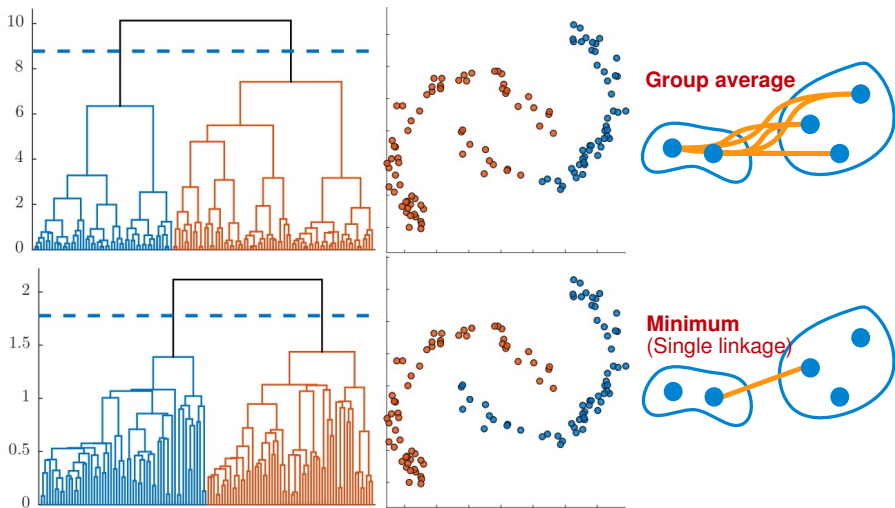
Solution:

The correct answer is *D*. The clusters will merge at height 5, 6, 12 and 25.

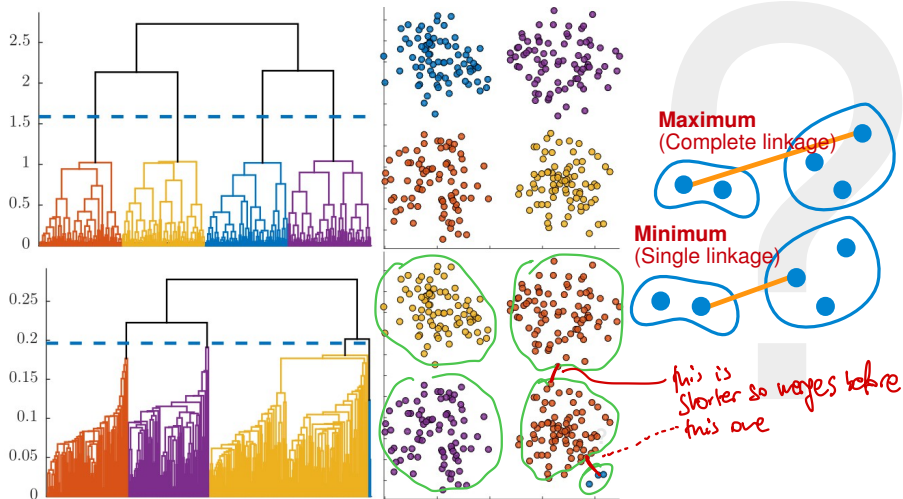
Clusterings and linkage functions



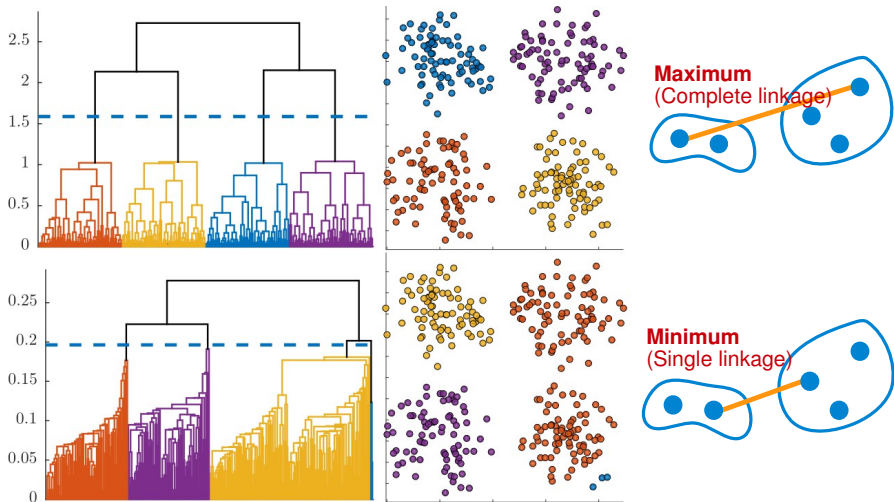
Clusterings and linkage functions



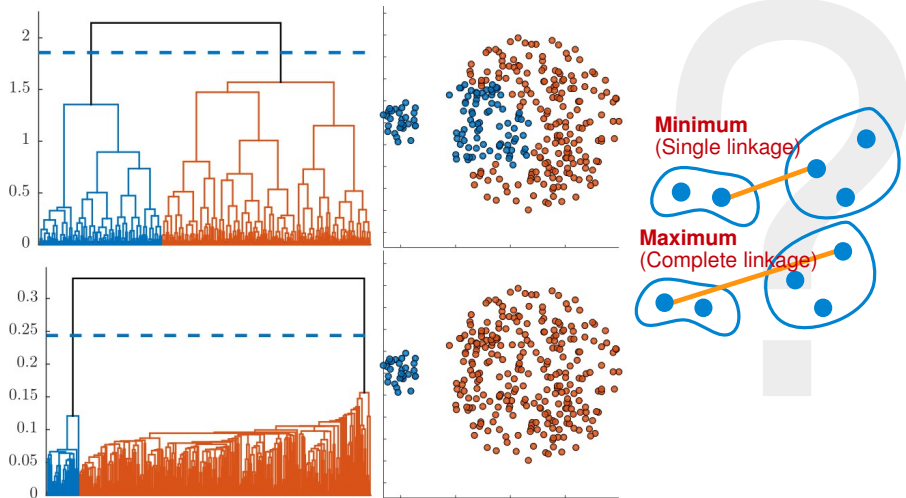
Clusterings and linkage functions



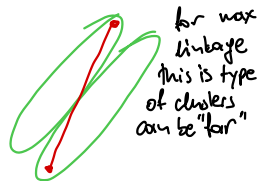
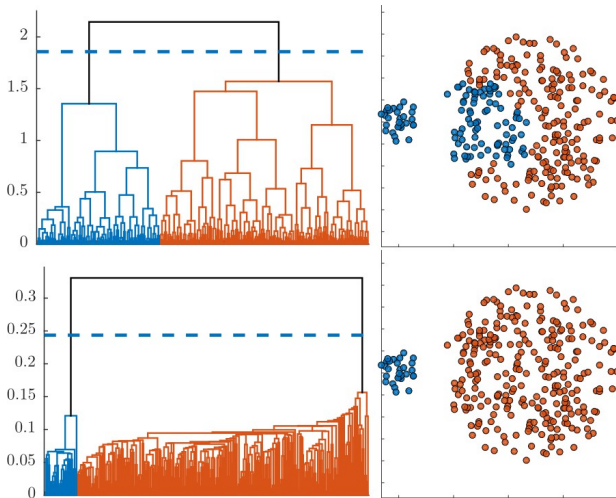
Clusterings and linkage functions



Clusterings and linkage functions



Clusterings and linkage functions



Maximum
(Complete linkage)

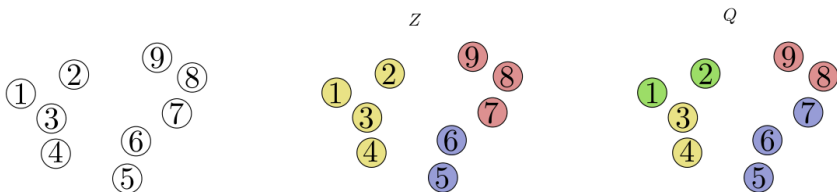


Minimum
(Single linkage)



Comparing partitions

- How similar are Q and Z



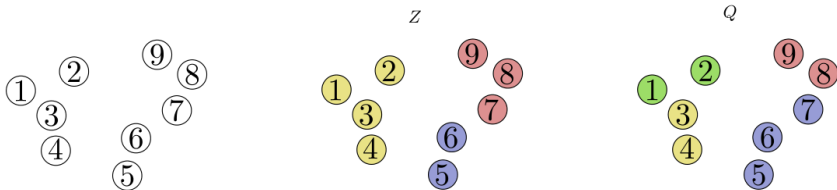
$$Z = [1 \quad 1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 3 \quad 3 \quad 3]$$

$$Q = [4 \quad 4 \quad 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 3 \quad 3]$$

- Note encoding is (and should be!) arbitrary

$$Q' = [10 \quad 10 \quad 3 \quad 3 \quad 8 \quad 8 \quad 8 \quad 1 \quad 1]$$

Comparing partitions



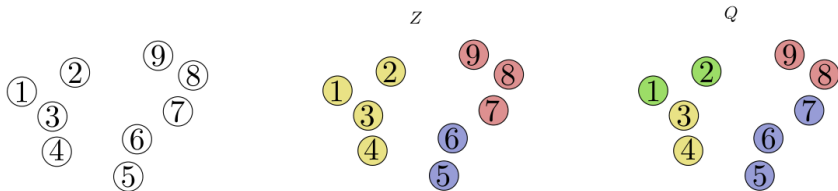
- For a given partition, any two observations i, j can either be in the same cluster, or in different clusters

$$\delta_{i,j} = \begin{cases} 0 & \text{if } o_i \text{ and } o_j \text{ are in different clusters} \\ 1 & \text{if } o_i \text{ and } o_j \text{ are in the same cluster} \end{cases}$$

- There are $\frac{1}{2}N(N-1)$ unique pairs in total
- We get two $\frac{1}{2}N(N-1)$ -long binary vectors corresponding to each pair (i, j)

$$\begin{aligned} \mathbf{b}^Z &= \begin{bmatrix} \delta_{1,2} & \delta_{1,3} & \delta_{1,4} & \delta_{1,5} & \cdots & \delta_{2,3} & \delta_{2,4} & \cdots & \delta_{3,5} & \cdots & \delta_{N,N-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 0 & \cdots & 1 & 1 & \cdots & 0 & \cdots & 1 \end{bmatrix} \\ \mathbf{b}^Q &= \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix} \end{aligned}$$

Comparing partitions - Jaccard and SMC/Rand index



$$\mathbf{b}^Z = \begin{bmatrix} 1 & 1 & 1 & 0 & \dots & 1 & 1 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{b}^Q = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix}$$

$S = \{ \text{Number of pairs in the same cluster in } Z \text{ and } Q \}$

$D = \{ \text{Number of pairs in different clusters in } Z \text{ and } Q \}$

$$\text{Rand index: } R(Z, Q) = \frac{S + D}{\frac{1}{2}N(N-1)} = \frac{4 + 24}{\frac{1}{2}9 \cdot 8} = \frac{7}{9},$$

$$\text{Jaccard similarity: } J(Z, Q) = \frac{S}{\frac{1}{2}N(N-1) - D} = \frac{4}{\frac{1}{2}9 \cdot 8 - 24} = \frac{1}{3}$$

An efficient encoding

$$n_{km} = \{\text{Observations assigned to cluster } k \text{ in } Z \text{ and } m \text{ in } Q\} = \sum_{i=1}^N \sum_{j=1}^N \delta_{z_i, k} \delta_{z_j, m}$$

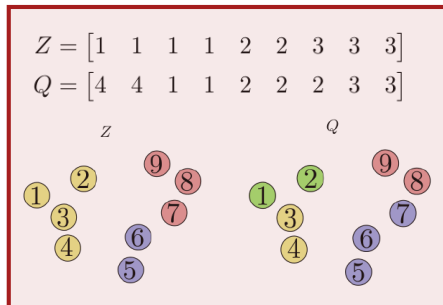
$$n^Z = \{\text{Number of observations assigned to cluster } k \text{ in } Z\} = \sum_{m=1}^M n_{km}$$

$$n^Q = \{\text{Number of observations assigned to cluster } m \text{ in } Q\} = \sum_{k=1}^K n_{km}$$

$$n = \begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix} \begin{matrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{matrix} \begin{matrix} Z \\ Q \end{matrix}$$

Note the horizontal/vertical sums of n :

$$n^Z = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}, \quad n^Q = \begin{bmatrix} 2 & 3 & 2 & 2 \end{bmatrix}$$



Jaccard and rand index in general

Recall

$$\mathbf{n} = \begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}, \quad \mathbf{n}^Z = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{n}^Q = [2 \quad 3 \quad 2 \quad 2]$$

$S = \{ \text{Number of pairs } i, j \text{ in the same cluster in } Z, Q \}$

$$\begin{aligned} &= \sum_{k=1}^K \sum_{m=1}^M \frac{n_{km}(n_{km} - 1)}{2} \\ &= \frac{2(2-1)}{2} + \frac{2(2-1)}{2} + \frac{2(2-1)}{2} + \frac{1(1-1)}{2} + \frac{2(2-1)}{2} = 4 \end{aligned}$$



$D = \{ \text{Number of pairs } i, j \text{ in different clusters in } Z, Q \}$

$$\begin{aligned} &= \frac{N(N-1)}{2} - \sum_{k=1}^K \frac{n_k^Z(n_k^Z - 1)}{2} - \sum_{m=1}^M \frac{n_m^Q(n_m^Q - 1)}{2} + S \\ &= 36 - 10 - 6 + 4 = 24 \end{aligned}$$



Q Quiz 04: Cluster overlap

		o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
C_1	o_1	0.0	2.0	5.7	0.9	2.9	1.8	2.7	3.7	5.3	5.1
	o_2	2.0	0.0	5.6	2.4	2.5	3.0	3.5	4.3	6.0	6.2
C_2	o_3	5.7	5.6	0.0	5.0	5.1	4.0	3.3	5.4	1.2	1.8
	o_4	0.9	2.4	5.0	0.0	2.7	2.1	2.2	3.5	4.6	4.4
	o_5	2.9	2.5	5.1	2.7	0.0	3.5	3.7	4.0	5.8	5.7
C_3	o_6	1.8	3.0	4.0	2.1	3.5	0.0	1.7	5.3	3.8	3.7
	o_7	2.7	3.5	3.3	2.2	3.7	1.7	0.0	4.2	3.1	3.2
	o_8	3.7	4.3	5.4	3.5	4.0	5.3	4.2	0.0	5.5	6.0
	o_9	5.3	6.0	1.2	4.6	5.8	3.8	3.1	5.5	0.0	2.1
	o_{10}	5.1	6.2	1.8	4.4	5.7	3.7	3.2	6.0	2.1	0.0

Table 1: The pairwise distances between $N = 10$ observations from the travel review dataset. the colors indicate classes

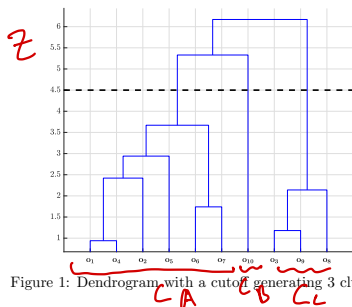


Figure 1: Dendrogram with a cutoff generating 3 clusters.

Consider the dendrogram in Figure 1. Suppose we apply a cutoff (indicated by the black line) thereby generating three clusters. We wish to compare the quality of this clustering, Q , to the ground-truth clustering, Z , indicated by the colors in Table 1. Recall the *Jaccard similarity* of the two clusters is

$$J[Z, Q] = \frac{S}{\frac{1}{2}N(N-1) - D}$$

in the notation of the lecture notes. What is the Jaccard similarity of the two clusterings?

- A. $J[Z, Q] \approx 0.104$
- B. $J[Z, Q] \approx 0.143$
- C. $J[Z, Q] \approx 0.174$
- D. $J[Z, Q] \approx 0.153$
- E. Don't know.

$$Q = \begin{bmatrix} C_1 & C_2 & C_3 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{bmatrix} \begin{matrix} C_A \\ C_B \\ C_C \end{matrix}$$

$$Z = [C_A, C_A, C_C, C_A, C_A, C_A, C_A, C_C, C_C, C_B]$$

$$Q = [C_1, C_1, C_2, C_2, C_2, C_3, C_3, C_3, C_3, C_3]$$

Solution:

To compute $J[Z, Q]$, note Z is the clustering corresponding to the colors in ?? and Q the clustering obtained by cutting the dendrogram in ?? given as:

$$\{10\}, \{1, 2, 4, 5, 6, 7\}, \{3, 8, 9\}$$

From this information we can define the counting matrix $\mathbf{n}$ as

$$\mathbf{n} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

It is then a simple matter of using the definitions in the lecture notes (see chapter 17.4) to compute

$$S = 4, D = 17$$

From this the answer by simply plugging the values into the formula given in the text and answer B is correct.

Entropy and mutual information recap

- Consider a probability distribution $P(X = x_i) = p_i, i = 1, \dots, n$
- Information** obtained from observing x_i is

$$I = -\log p_i$$

- Average information obtained is called the **entropy**

$$H[p_X] = \mathbb{E}[I] = -\sum_{i=1}^n p_i \log p_i$$

- Entropy is defined for general densities $P(X = x_i, Y = y_j) = p_{ij}$

$$H[p_{XY}] = -\sum_{i=1}^n \sum_{j=1}^m p_{ij} \log p_{ij}$$

- The **Mutual information** is defined as

$$\text{MI}[X, Y] = H[P_X] + H[P_Y] - H[P_{XY}]$$

- The **Normalized mutual information** is defined as

$$\text{NMI}[X, Y] = \frac{\text{MI}[X, Y]}{\sqrt{H[P_X]} \sqrt{H[P_Y]}}$$

Comparing using mutual information

We define $P_{ZQ}(i, j) = \frac{1}{N}n_{ij}$, $P_Z(i) = \frac{n_i^Z}{N}$ and $P_Q(j) = \frac{n_j^Q}{N}$. Example:

$$P_{ZQ} = \frac{1}{9} \begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}, P_Z = \frac{1}{9} \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}, P_Q = \frac{1}{9} [2 \quad 3 \quad 2 \quad 2]$$

- **Entropy** computed as $H[p_X] = -\sum_{i=1}^n p_i \log p_i$:

$$\text{Entropy of } Z: H[Z] = -\frac{4}{9} \log \frac{4}{9} - \frac{1}{3} \log \frac{1}{3} - \frac{2}{9} \log \frac{2}{9} \approx 1.06$$

$$\text{Entropy of } Q: H[Q] = -\frac{2}{9} \log \frac{2}{9} - \frac{2}{9} \log \frac{2}{9} - \frac{2}{9} \log \frac{2}{9} - \frac{1}{3} \log \frac{1}{3} \approx 1.37$$

$$\text{Entropy of } Z \text{ and } Q: H[ZQ] = -4 \times \frac{2}{9} \log \frac{2}{9} - \frac{1}{9} \log \frac{1}{9} = 1.58.$$

- **Mutual information:**

$$\text{MI}[Z, Q] = H[Z] + H[Q] - H[Z, Q] \approx 1.06 + 1.37 - 1.58 \approx 0.85.$$

- **Normalized mutual information:**

$$\text{NMI}[Z, Q] = \frac{\text{MI}[Z, Q]}{\sqrt{H[Z]}\sqrt{H[Q]}} \approx \frac{0.85}{\sqrt{1.06}\sqrt{1.37}} \approx 0.70.$$

Summary

- **Clustering**

Idea: *Group observations that are similar in some way*

- K-means

Key aspects: K, initialization, distance function and method for recomputing centroids

- Hierarchical clustering and linkage functions

Key aspects: distance function, linkage function (minimum/single, maximum/complete, average and Ward), and K/cutoff threshold

- **Comparison of clustering solutions**

Idea: *Consider which observations are clustered together (or not) in the two solutions*

- Rand index
 - Jaccard
 - NMI

Resources